Factors Affecting Consumer Food Price in India: An Econometric Analysis

Shambhavi Mittal
City Montessori School
Cambridge Section, Gomti Nagar, Lucknow 226010
shambhavimittal14@gmail.com

Abstract

This study examines the determinants of food price movements in India with the aim of assessing the influence of exchange rates, crude oil prices, monetary policy, and fertilizer prices on the Consumer Food Price Index (CFPI). The research focuses on the relationship between CFPI and five key macroeconomic and input-cost variable which are USD/INR exchange rate, EUR/INR exchange rate, Brent crude oil prices, the repo rate, and the Wholesale Price Index (WPI) of fertilizers and nitrogen compounds within the Indian economic context. The study adopts a quantitative secondary-data time-series econometric design using annual data for the period 2000-2025. The data was analyzed using descriptive statistics, Pearson pairwise correlation, multiple linear regression and multicollinearity diagnostics in STATA to evaluate the relationship between food prices and the selected explanatory variables.Model 1 included all five explanatory variables, while Model 2 retained Fertilizer WPI, USD/INR and EUR/INR following the multicollinearity assessment. The final regression results indicate that fertilizer WPI has a coefficient of 1.650269 and a P-value of 0.000 and EUR/INR exchange rate has a coefficient of 1.239716 with a P-value of 0.000 are statistically significant at 5%, while USD/INR is significant of 10% level. The final model reports an R-squared of 0.9847. These findings indicate statistical associations between selected agricultural input-cost and macroeconomic variables and CFPI within the specified model, rather than establishing causal relationships. The study contributes to the literature by examining agricultural input costs alongside selected macroeconomic variables within a common empirical framework.

Keywords: Consumer Food Price Index (CFPI), Fertilizer Prices, Food Price Inflation, Exchange Rates, Brent Crude Oil, Repo Rate

Introduction

Background of the study:

In India, the Consumer Food Price Index(CFPI) measures the changes in the retail prices of food items consumed by households and therefore provides an appropriate indicator for examining movement in consumer food prices. While food inflation refers to the rate of change in food prices, this study uses CFPI level as the dependent variable and examines consumer food price movement rather than the food inflation rate.

Food prices are an essential component of India’s overall inflation dynamics because change in prices of essential goods and services directly affect the purchasing power and living costs of households. Food price volatility adversely affects low-income households by eroding their purchasing power, as they spend a significant portion of their income on food. This volatility creates uncertainty, making it difficult for these households to maintain adequate nutrition.(Chandrasekhar, 2013). In India, the Consumer Food Price Index(CFPI) measures changes in the retails prices of food items consumed by households, hence, is an appropriate indicator for examining food price movements.

© 2026 The Author(s). Published in Global Confluence Review (ISSN 3139-6690) under the CC BY 4.0 licence: https://creativecommons.org/licenses/by/4.0/
Available online: https://www.globalconfluencereview.in/issues/vol-1-issue-2-2026

Food prices are influenced by both domestic and international factors. Changes in crude oil prices can affect transportation and agricultural input costs, while the exchange rate movements can influence the domestic prices of imported goods and services, which are often used as inputs like fertilizers and petroleum products. Previous research on India has found relationships between food inflation and variables including exchange rates, crude oil prices, interest rates and global food prices (Ahmed & Singla, 2014; Samal et al., 2022).

Fertilizers constitute an important agricultural input, hence, are considered particularly relevant, an increase in input costs can raise the cost of agricultural production, leading to an increase in food price as well. Similarly, international crude oil prices also affect the food prices by contributing transportation, energy and other production costs. Additionally exchange rate movements can also influence domestic prices by altering the rupee cost of imported goods and services.

Previous empirical studies have consequently examined food prices and food inflation through a range of macroeconomic and supply-side determinants. Ahmed and Singla (2014) analysed Indian data and found relationships between food inflation and factors including interest rates and exchange rates. Similarly, Samal, Ummalla, and Goyari (2022) found that macroeconomic factors, including exchange rates and global food prices, are associated with food-price movements in India. More recent research has also emphasised crude oil prices, international food prices, monetary conditions, and other supply-side factors.

Problem Statement and Rationale:

Food price movements in India are influenced by a combination of domestic and international economic factors. Previous researchers have established the food inflation can be affected by several factors like global commodity prices and monetary conditions, however the relative importance of these factors may vary across time and economic conditions. Existing empirical studies have employed different datasets, time periods, frequencies and methodological approaches, which makes it difficult to establish a consistent conclusion. A particular area which requires more investigation is the role of agricultural input costs alongside macroeconomic variables. Factors such as fertilizer prices, crude oil prices, exchange rates and monetary policy may influence food prices through different transmission mechanism, yet their relationships with consumer food prices may not be unfirm. Therefore, the problem addressed by this study is the need to assess the extent to which selected macroeconomic and agricultural input cost variables explain variations in India's Consumer Food Price Index. These variables are examined with a common regression framework over the the period of 2000-2025- the study seeks to identify which factors demonstrate statistically significant relationships with food prices in India.

Research Objectives

The primary objective of this study is to empirically examine the factors affecting food prices in India by analysing the relationship between Consumer Food Price Index (CFPI) and selected macroeconomic and agricultural input-cost variables over the period of 2000-2025.

Significance of the study

Food price volatility in India significantly reduces the purchasing power of low-income households, as uncertainty in prices acts as a barrier to escaping poverty, leading to malnutrition, decreased economic welfare, and potential withdrawal of children from education. (Wibowo et al., 2023). This study identifies the factors associated with changes in food prices, which can contribute to a better understanding od the

channels through which domestic and international economic conditions affect consumers and agricultural markets.

The study may be relevant to policymakers because identifying statistically significant associations between selected variables and CFPI can contribute to a better understanding of the factors associated with consumer food prices. Agricultural policymakers may also find the findings relevant in considering fertilizer and other agricultural input-cost pressures. The findings should, however, be interpreted as associations within the selected model rather than as evidence that any individual factor causes changes in CFPI.

The study also contributes to academic and economic research by examining exchange rates, crude oil prices, monetary policy and fertilizer prices simultaneously within a single empirical framework, providing a broader perspective than examining individual determinants in isolation.

Lastly, this study may provide a basis for future researchers to extend the analysis by incorporating additional factors such as rainfall, agricultural productions and many more.

Scope and Delimitations

This study focuses on factors associated with consumer food prices in India, with the Consumer Food Price Index (CFPI) serving as the dependent variable. The analysis covers the period from 2000 to 2025 and uses annual secondary data obtained from official government and authoritative institutional sources. The scope is limited to the selected variables and does not provide an exhaustive explanation of all factors associated with food prices in India. Factors such as rainfall, temperature, agricultural production, crop yields, minimum support prices, food stocks, imports and exports, wages, money supply, government subsidies, and international food commodity prices are outside the primary scope of the model.

The scope of the study is limited to the selected variables and does not provide an exhaustive explanation of all determinants of food prices in India. Factors such as rainfall, temperature, agricultural production, crop yields, minimum support prices, food stocks, imports and exports, wages, money supply, government subsidies, and international food commodity prices are outside the primary scope of the model.

Methodology

This study employed a quantitative secondary-data time-series econometric research design to investigate the relationship between selected macroeconomic and agricultural input cost variables and the Consumer Food Price Index (CFPI). The research was conducted in India, chosen because food price movements have significant effects on inflation, household purchasing power, agricultural production and economic stability. The study used secondary data, data obtained from official government of India databases, including sources for Consumer Food Price Index (CFPI), exchange rates, The Reserve Bank of India's repo rate, fertilizer WPI and Brent Crude Oil Prices. Data were gathered through the compilation of published annual secondary datasets from relevant official sources. To ensure the validity and reliability of the research instruments, the study relied on data obtained from official government sources . The collected data were analysed using STATA statistical software employing techniques such as descriptive statistics, correlation analysis, multiple linear regression, Variance Inflation Factor (VIF), heteroskedasticity testing for quantitative interpretation. Ethical considerations such as accurate representation of secondary data, appropriate acknowledgement of data sources, transparency in data processing, and responsible reporting of findings were strictly observed, with approval obtained from the relevant academic/research supervisor or institution. Although the study faced limitations including the relatively small number of annual

observations, limited availability of historical data the chosen methodology was appropriate for addressing the research objectives.

Literature Review:

Food price volatility has important welfare implications in India, particularly for lower-income households that allocate a relatively large proportion of their income to food. Increased food-price uncertainty can reduce purchasing power and create difficulties in maintaining adequate nutrition (Chandrasekhar, 2013). Similarly, Wibowo et al. (2023) find that food-price volatility can reduce household economic welfare and create adverse consequences for lower-income households. Bhattarai et al. (2025) further show that increases in food prices can generate heterogeneous consumption responses across income groups, with lower-income households experiencing relatively greater adverse effects.

The literature identifies several channels through which food prices may change, including agricultural input costs, energy prices, exchange rates, monetary conditions, rainfall, storage constraints and global commodity-market conditions. Ahmed and Singla (2014), for example, identify world food prices and crude oil prices as important influences on food inflation in the short run. Diwakar (2025) additionally highlights seasonal cycles, inadequate storage infrastructure, post-harvest losses, government policies and global market conditions as factors contributing to food-price fluctuations, particularly for perishable commodities. These findings suggest that consumer food prices reflect multiple transmission mechanisms rather than a single determinant. The present study therefore focuses on four broad channels represented by its explanatory variables: agricultural input costs, energy costs, exchange-rate movements and monetary policy.

Fertilizer prices represent an important agricultural input-cost channel through which changes in production costs may be transmitted to food prices. Calvia (2025) identifies fertilizer prices as significant drivers of food prices, with higher fertilizer costs potentially increasing the cost of agricultural production and subsequently affecting consumer prices. Kun and Takács (2023) similarly identify a relationship between fertilizer prices, energy costs and movements in the Consumer Food Price Index (CFPI).

The relationship between fertilizer prices and food prices may, however, depend on the broader structure of agricultural and global commodity markets. Alexander et al. (2022) associate higher fertilizer prices, particularly when driven by energy-price shocks, with increased agricultural input costs and pressures on food affordability. Alviarez et al. (2025) further find that fertilizer-price shocks can affect crop prices, although the degree of pass-through differs according to the source of the price shock. Conversely, Yakimovich (2023) argues that although fertilizer prices increase production costs, they are not necessarily the primary driver of food inflation, as financial and structural characteristics of the global food system can also influence food prices. The *Inorganic Fertilizer Price Surge in 2021: Key Drivers and Policy Options* (2022) also discusses the relationship between fertilizer demand, fertilizer prices and food prices.

Overall, the literature supports fertilizer prices as a relevant cost-side transmission channel, while also indicating that the relationship may not be uniform across markets or periods. This provides a basis for including Fertilizer WPI as an explanatory variable in the present model.

Crude oil prices provide a second cost-side transmission channel because changes in energy costs can affect transportation, agricultural production and other stages of the supply chain. Choi et al. (2017) find that a 10% increase in global oil prices is associated with an increase in domestic inflation of approximately 0.4 percentage points on impact, although the effect diminishes over time. Chen and Yang (2024) similarly

identify gasoline and fuel oil as important contributors to CPI inflation and report substantial volatility in energy commodities, including Brent crude oil.

Other research also identifies a relationship between oil-price movements and inflation. Chen (2008) finds that oil-price fluctuations affect inflation, although the magnitude of the pass-through differs across categories and has declined over time. Renou-Maissant (2019) reports evidence of oil-price effects on inflation dynamics across countries, while Prabha et al. (2023) highlight the importance of oil-price movements for domestic inflation. Adebayo (2020) identifies positive co-movement between oil prices and inflation and reports evidence of a directional relationship from oil prices to inflation during the period examined. Fukač (2011) similarly examines the pass-through from oil prices to inflation, while Alp et al. (2023) associate the sharp increase in Brent crude oil prices in 2022 with increased inflationary pressure.

Although much of this literature examines overall CPI inflation rather than CFPI specifically, it provides a theoretical basis for considering crude oil as an indirect input-cost channel for consumer food prices. In the present study, Brent crude oil prices are therefore included in the initial model to examine whether this broader energy-cost relationship is also reflected in consumer food prices in India.

Exchange-rate movements constitute another potential transmission mechanism because changes in the domestic currency value of imported goods and inputs can affect domestic prices. Goldberg and Campa (2010) demonstrate that changes in exchange rates can influence consumer prices through imported input costs, although the magnitude of exchange-rate pass-through differs substantially across countries. Liu (2023) similarly discusses the relationship between movements in the US dollar and inflationary pressures, particularly in emerging and non-US economies. Mohamed and Saleh (2024) further identify exchange-rate fluctuations as an important component of inflation dynamics through exchange-rate pass-through and international price transmission.

Within the present framework, both USD/INR and EUR/INR are included because exchange-rate exposure is not restricted to a single international currency. The US dollar is particularly relevant to international commodity and energy markets, while the euro represents another major international currency through which trade and imported goods can be priced. Including both exchange rates allows the model to examine whether CFPI is associated with broader exchange-rate movements rather than assuming that movements in the US dollar alone capture all relevant external price transmission.

However, the inclusion of both variables also creates an econometric consideration: exchange rates may themselves move together because they respond to common international and domestic macroeconomic conditions. Therefore, the present study evaluates multicollinearity using the Variance Inflation Factor (VIF) and considers the implications of correlated exchange-rate variables when specifying the final regression model.

Monetary policy provides a demand-side transmission channel that differs from the cost-side mechanisms represented by fertilizer and crude oil prices. Changes in the Reserve Bank of India's repo rate can influence borrowing costs, consumption, investment and aggregate demand, thereby affecting inflationary pressures. *Analyzing the Correlation between Repo Rate and Macroeconomic Stability in India (2010–2024)* (2025) reports a relationship between repo-rate movements and CPI inflation, while Varshini et al. (2024) similarly identify a linear relationship between the repo rate and inflation, while recognising the influence of other economic factors.

The effectiveness of monetary policy may nevertheless depend on the source of inflationary pressure. Suresh (2012) argues that repo-rate adjustments may have more limited effects when inflation is primarily supply-driven rather than demand-driven. Similarly, *An Investigation into Impact of Repo Rate on Inflation Indices (WPI & CPI) in India during Pandemic and Post Pandemic Phase* reports a relatively limited relationship between repo-rate changes and inflation. Kumar et al. (2010), in contrast, report a moderate-to-strong positive correlation between the repo rate and CPI inflation and suggest that gradual changes in the rate may be more effective in moderating inflation.

These findings indicate that the repo rate may operate differently from fertilizer, oil and exchange-rate variables because it represents a policy response and demand-side mechanism, rather than a direct production or imported-cost determinant. Its inclusion in the initial model therefore allows the study to examine whether monetary conditions are statistically associated with CFPI alongside supply-side and external variables.

Existing literature provides evidence that Indian food prices are associated with multiple economic mechanisms, including agricultural input costs, energy prices, exchange-rate movements and monetary conditions. However, much of the existing research examines these channels separately, focuses on overall CPI or food inflation, or uses different periods and methodological approaches. Consequently, there is scope for a framework that considers several transmission channels simultaneously using a common measure of consumer food prices.

The present study contributes to this literature by examining the Consumer Food Price Index (CFPI) in India using a common annual dataset covering 2000–2025 and by bringing together four theoretically distinct channels within the same empirical framework: fertilizer prices as an agricultural input-cost channel, Brent crude oil as an energy-cost channel, USD/INR and EUR/INR as exchange-rate channels, and the RBI repo rate as a monetary-policy channel. The study first estimates a model containing all five explanatory variables and then evaluates multicollinearity before specifying a reduced model.

A further contribution is the explicit consideration of both USD/INR and EUR/INR rather than treating exchange-rate exposure as a single variable. This allows the study to examine whether consumer food prices are associated with movements in different major international currencies while also testing whether their simultaneous inclusion creates multicollinearity. Rather than claiming that these variables cause changes in CFPI, the study evaluates their statistical associations and compares the relative relationships across the proposed transmission channels.

Thus, the study extends earlier literature by providing an integrated empirical examination of selected agricultural, energy, exchange-rate and monetary-policy channels associated with consumer food prices in India over a common 2000–2025 period.

Methods

Research Design:

This study utilises a quantitative secondary-data time-series econometric research design to examine associations between CFPI and selected macroeconomic and agricultural input-cost variables in India. The analysis covers annual observations from 2000–2025, resulting in 26 observations.

Participants or Sample (Data Sources)

A specific multi-decade timeline from 2000-2025 is issued, providing 26 annual observations for examining the factors influencing food price movements. The use of a multi-decade time series enables the study to capture changes in food prices and their relationship with specific independent variables including fertilizer prices, The Reserve Bank of India's repo rate and other relevant economic indicators, including periods of significant economic disruption. The data were obtained primarily from official government and institutional sources, including Reserve Bank of India and relevant government of India statistical databases. The data remains credible as it was taken from verified government and institution frameworks, and is not likely to be prone to research bias.

Variables and Measurements:

Dependent Variable: Consumer Food Price Index (CFPI)

Independent Variable: Fertilizer Prices, The Reserve Bank of India's repo rate, Brent Oil Prices, USD/INR exchange rate, EUR/INR exchange rate, Continuous Time-Trend Variable (Year).

The continuous time-trend variable (Year) is used to capture changes over time

Data Analysis:

A Pearson pairwise correlation matrix was built to understand the linear relationships between the variables and observe if there are any potential multicollinearity issues amongst the independent variables, which is when the independent variables are highly correlated with each other, which can lead to less precise interpretations. If it is present, the redundant independent variables can be removed from the model.

Two regression models were employed. The first model incorporated all the selected independent variables, while the second model excluded variables exhibiting multicollinearity in order to obtain more reliable estimates of the individual determinants of food-price movements in India. The Variance Inflation Factor (VIF) was used to assess multicollinearity among the explanatory variables, ensuring that highly correlated predictors did not distort the estimated coefficients.

Analysis and Results:

Regression analysis has been utilised to understand the effect of Fertilizer Prices, The Reserve Bank of India's repo rate, Brent Oil Prices, USD/INR exchange rate, EUR/INR exchange rate on Consumer Food Price Index. Linear regression models were examined to assess the linear relationship between dependent variable and each of the independent variable. Visual inspection indicated that EUR/INR exchange rate demonstrated a moderately positive relationship with CFPI. Similarly, fertilizer WPI and CFPI also exhibited a strong positive linear relationship along with Brent Oil Prices having a moderately strong association with CFPI. Moreover, USD/INR exchange rate also demonstrated a strong positive linear relationship. In contrast, a strong negative relationship was observed between repo rate and CFPI.

Figure 1 illustrates the regression model between CFPI and EUR/INR exchange rate. A positive upward-sloping linear relationship is observed between these two variables.

Figure 1: CFPI vs EUR/INR exchange rates

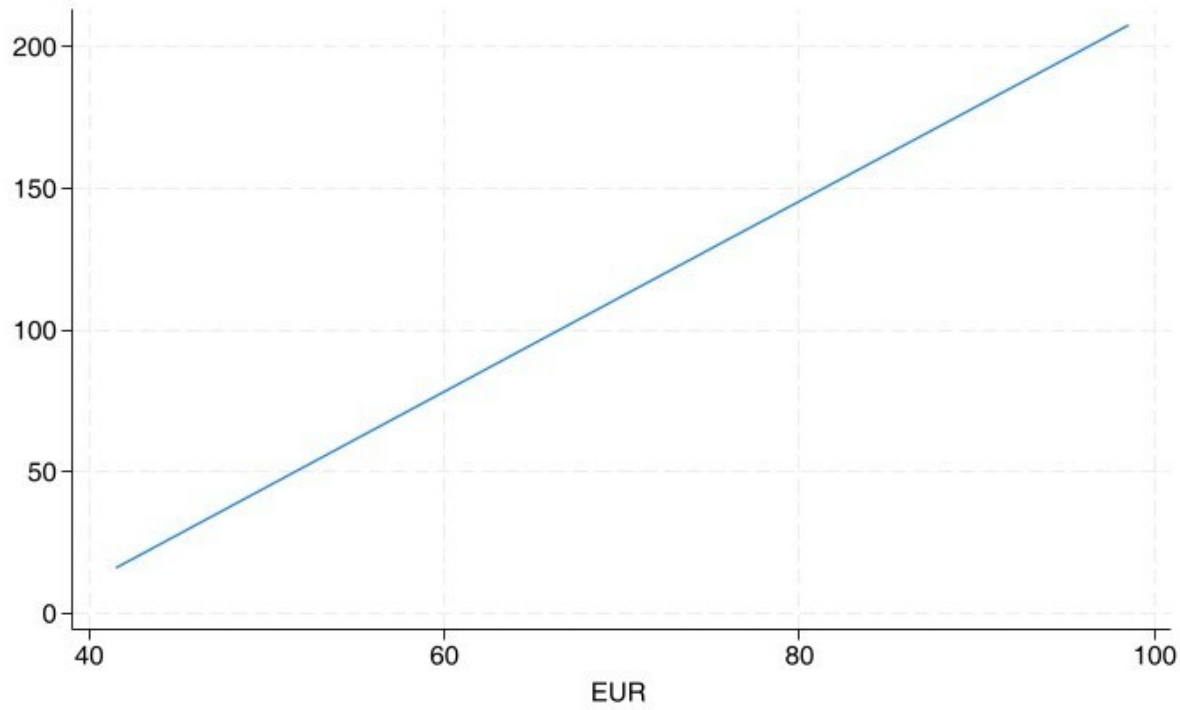


Figure 2: CFPI vs USD/INR exchange rates

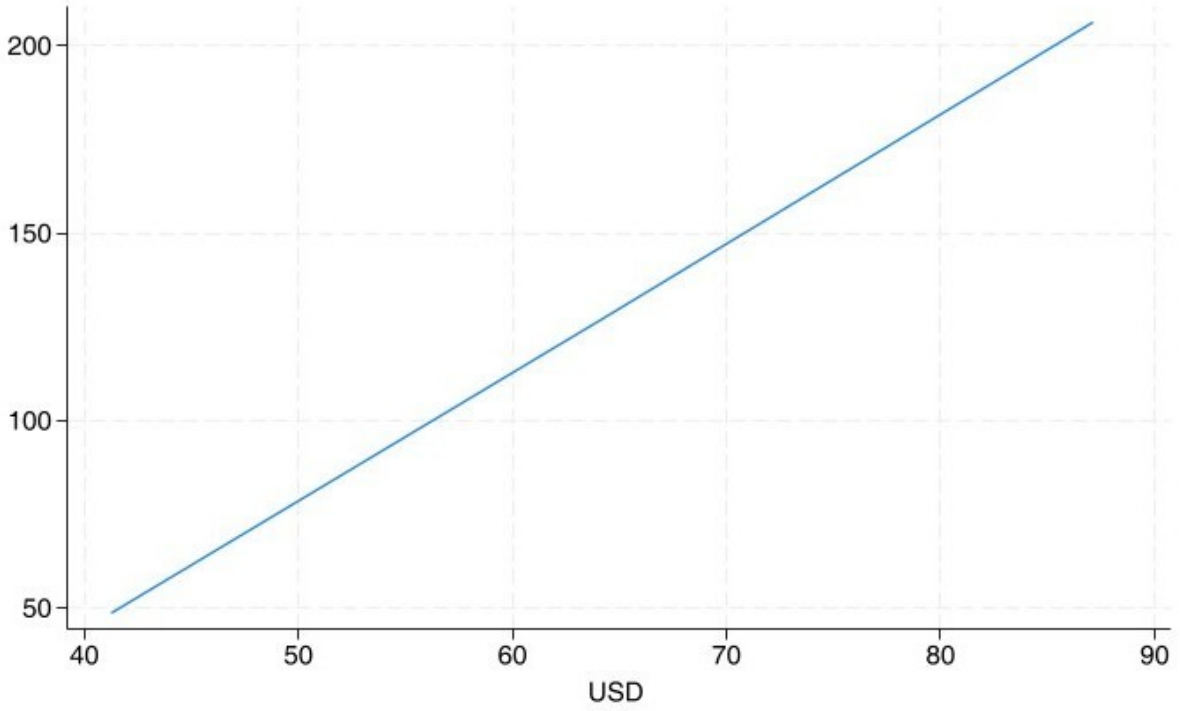


Figure 2 illustrates the regression model between CFPI and USD/INR exchange rate. A positive upward-sloping linear relationship is observed between these two variables.

Figure 3: CFPI vs Repo Rate

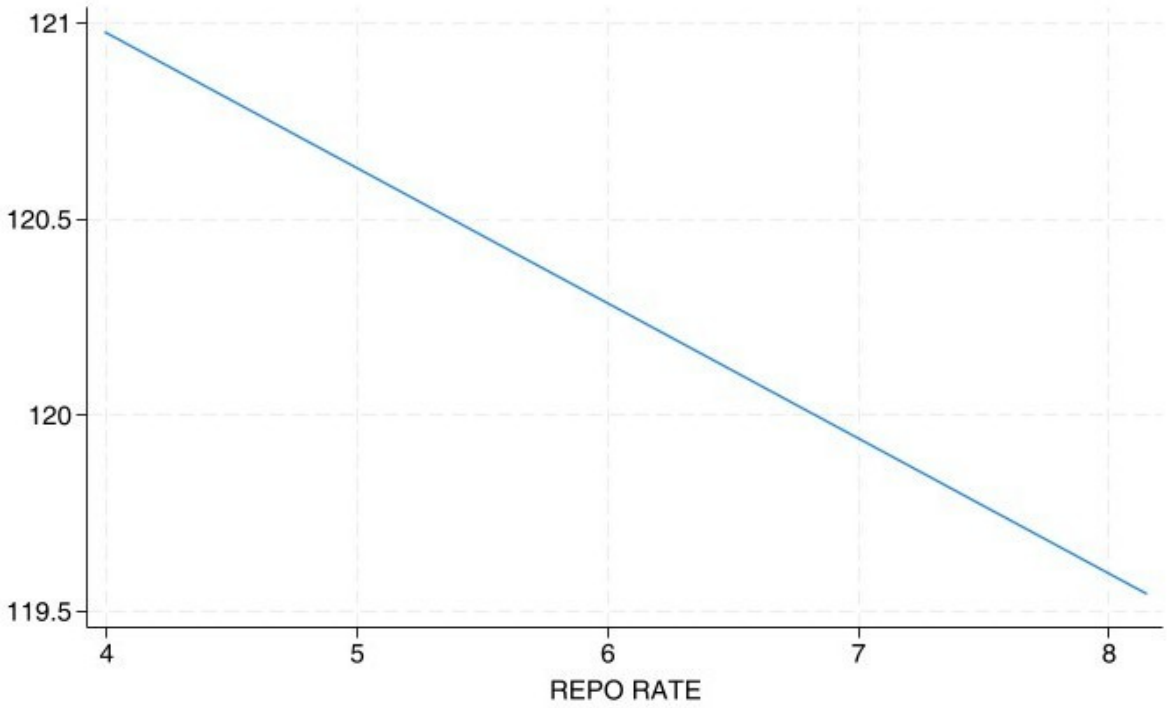


Figure 3 illustrates the regression model between CFPI and Repo Rate. A negative downward-sloping linear relationship is observed between these two variables.

Figure 4: CFPI vs Brent Oil Prices

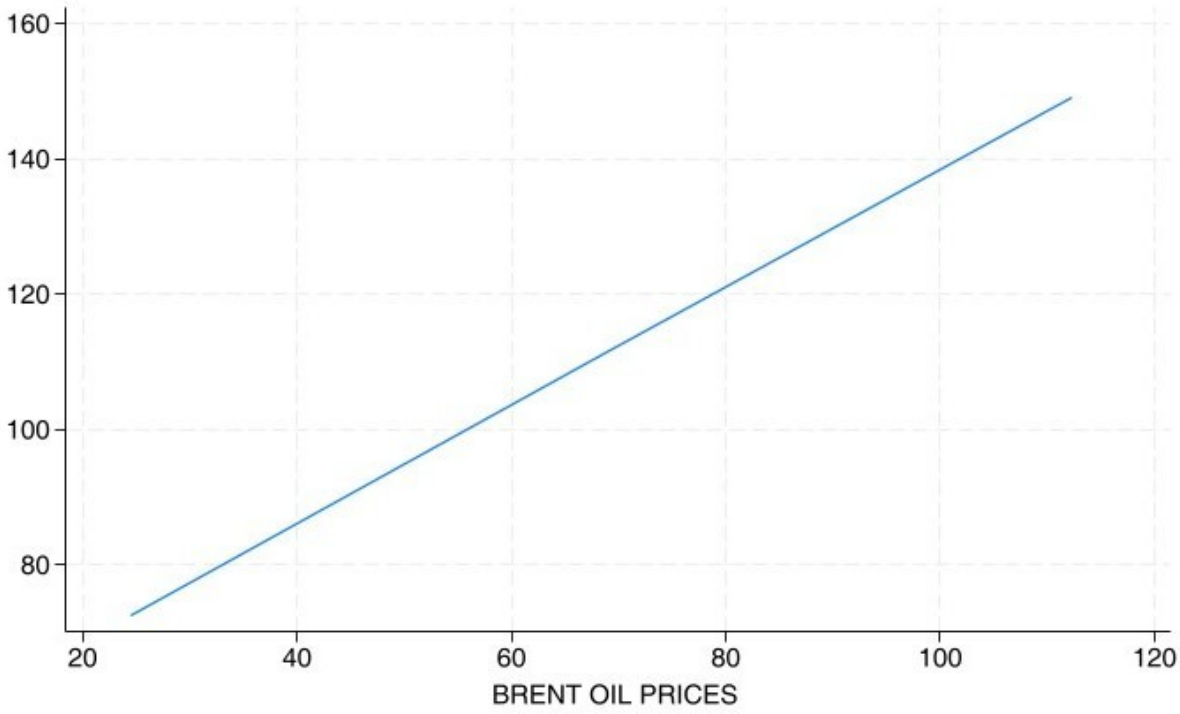


Figure 4 illustrates the regression model between CFPI and Brent Oil Prices. A positive upward-sloping linear relationship is observed between these two variables.

Figure 5: CFPI vs Fertilizer WPI

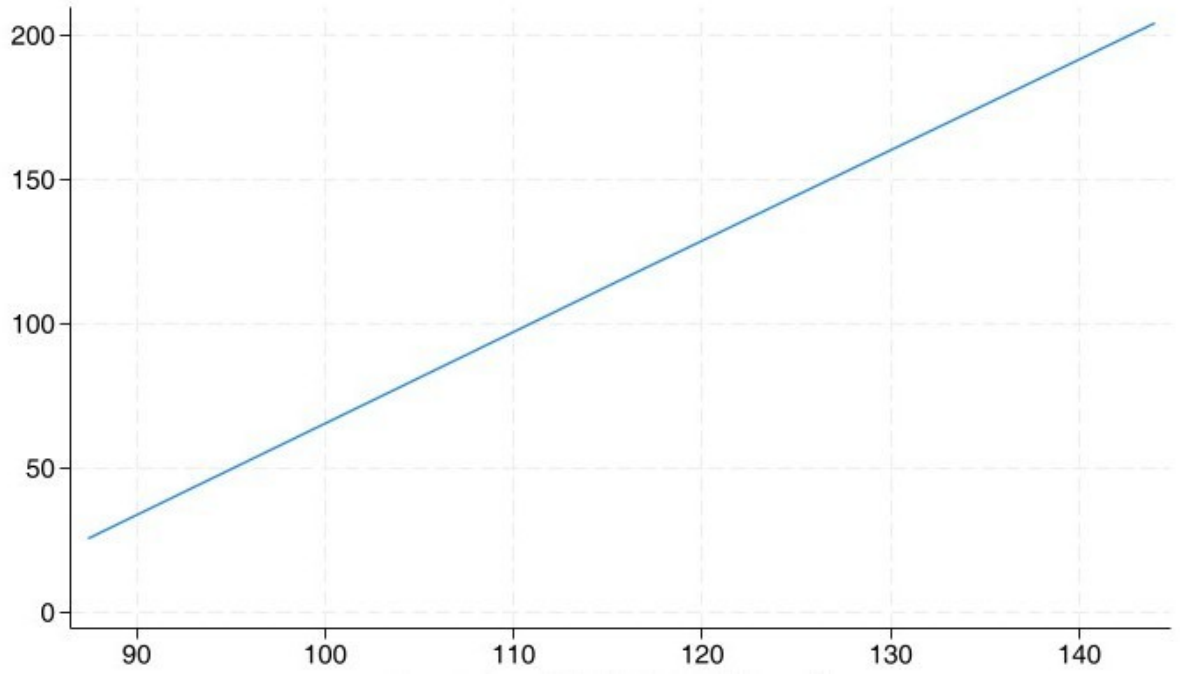
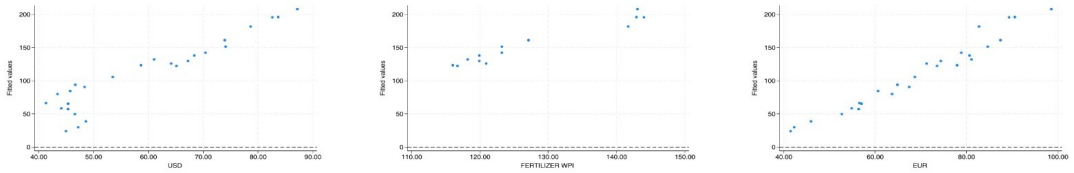


Figure 5 illustrates the regression model between CFPI and Fertilizer WPI. A positive upward-sloping linear relationship is observed between these two variables.

Figure 6: Residuals vs fitted plot



The residual-versus-fitted plot is used to assess whether the residuals display systematic patterns or changes in spread across fitted values. The plot should be interpreted in terms of the distribution of residuals rather than as evidence of a positive or negative relationship between the explanatory variables and CFPI.

The figure above shows residuals vs fitted plot for the variables Fertilizer WPI, EUR/INR Exchange Rate and USD/INR Exchange Rate. The figure illustrates an overall strong positive relationship of all variables with fitted values of CFPI.

ANOVA Table:

Source	**SS**	**df**	**MS**
Model	71206.6837	3	23735.5612
Residual	1106.17728	22	50.2807855
Total	72312.861	25	2892.51444

The R- squared is 71206.6837/ 72312.861 is 0.9847, showing that 98.47% of the variation in CFPI is explained by the independent variables.

Correlation Matrix Outcomes:

Figure 7: Correlation between CFPI and Brent Oil Prices

	cfpi_e~d	BRENTO~S
cfpi_extra~d	**1.0000**	
BRENTOILPR~S	**0.4352**	**1.0000**

Pearson correlation between CFPI and Brent Oil Prices is positive, with r = 0.4352, indicating a moderate positive bivariate association. This correlation does not by itself establish statistical significance in the multivariate regression model.

Figure 8: Correlation between CFPI and EUR/INR Exchange Rate

	cfpi_e~d	EUR
cfpi_extra~d	**1.0000**	
EUR	**0.9721**	**1.0000**

Pearson correlation between CFPI and EUR/INR Exchange Rate is positive, with r = 0.9721, indicating a very strong positive bivariate association.

Figure 9: Correlation between CFPI and USD/INR Exchange Rate

	cfpi_e~d	USD
cfpi_extra~d	1.0000	
USD	0.9361	1.0000

Pearson Correlation between CFPI and USD/INR Exchange Rate is positive, as r = 0.9361. This indicates a very strong positive relationship between the two variables.

Figure 10: Correlation between CFPI and Repo Rate

	cfpi_e~d	REPORATE
cfpi_extra~d	1.0000	
REPORATE	-0.0096	1.0000

Pearson Correlation between CFPI and Repo Rate is negative as r= -0.0096. This indicates that there is essentially no linear correlation.

Figure 11: Correlation between CFPI and Fertilizer WPI

	cfpi_e~d	fertil~d
cfpi_extra~d	1.0000	
fertilizer~d	0.9836	1.0000

Pearson Correlation between CFPI and Fertilizer WPI is positive as r = 0.9836. This indicates a very strong positive correlation between the two variables.

Regression Model:

Number of obs	=	23
F(5, 17)	=	157.26
Prob > F	=	0.0000
R-squared	=	0.9788
Adj R-squared	=	0.9726
Root MSE	=	7.9017

Figure 12: Multiple Linear Regression Model Summary for Model 1

Model 1 included all five explanatory variables and produced a VIF value of 21.72, indicating substantial multicollinearity among the explanatory variables. Following this assessment, Brent crude oil prices and the repo rate were excluded, resulting in Model 2, which contained Fertilizer WPI, USD/INR and EUR/INR. Model 2 produced a lower VIF value of 10.58, indicating that multicollinearity was reduced, although some degree remained. Therefore, Model 2 was retained as the final specification, and its regression results are

presented below. In the final Model 2 regression, 2 out of 3 variables are statistically significant at the 5% level. Fertilizer WPI has a coefficient of 1.650269 with a p-value of 0.000, while EUR/INR has a coefficient of 1.239716 with a p-value of 0.000. USD/INR has a coefficient of 0.5034184 and a p-value of 0.063. These results should be interpreted as statistical associations within the selected specification rather than causal effects.

cfpi	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
fertilize wpi	1.671524	.6017952	2.78	0.013	.4018472	2.941201
BRENTOILPRICES	-.0381921	.1625942	-0.23	0.817	-.3812358	.3048516
REPORATE	.1373774	1.735946	0.08	0.938	-3.525149	3.799903
EUR	1.100298	.5368652	2.05	0.056	-.0323887	2.232984
USD	.584964	.8413239	0.79	0.496	-1.190074	2.360002
_cons	-189.2993	33.36936	-5.67	0.000	-259.7025	-118.8961

Figure 13: Multiple Linear Regression Coefficients and Statistical Significance for Model 1

Number of obs	=	**26**
F(3, 22)	=	**472.06**
Prob > F	=	**0.0000**
R-squared	=	**0.9847**
Adj R-squared	=	**0.9826**
Root MSE	=	**7.0909**

Figure 14: Multiple Linear Regression Model Summary for Model 2

cfpi	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
fertilizer_wpi	1.650269	.3178884	5.19	0.000	.9910092	2.30953
USD	.5034184	.2571596	1.96	0.063	-.029898	1.036735
EUR	1.239716	.2948743	4.20	0.000	.628184	1.851248
_cons	-194.1759	15.06599	-12.89	0.000	-225.4208	-162.9309

Figure 15: Multiple Linear Regression Coefficients and Statistical Significance for Model 2

Discussion

Restatement of Key Findings:

The purpose of this study is to understand the relationship between CFPI of India and the independent variables examined (Fertilizer WPI, EUR/INR exchange rate, USD/INR Exchange Rate, Repo Rate and Brent Oil Prices). The key themes that have been identified include various degrees of significance of relationships with different variables. There is a positive significant relationship between CFPI and fertilizer WPI as fertilizer is a crucial agricultural input. When fertilizer prices increase, the farmers face higher production costs which leads to a significant rise in CFPI as food prices tend to increase. Moreover, a positive significant relationship between CFPI and EUR/INR exchange rate can also be observed. This relationship reflects the effect of international prices on imported goods and services which are most likely to be inputs for food production, directly affecting CFPI. Similarly, CFPI and Brent Oil Prices have a positive relationship but the correlation is weaker than the other selected independent variables. This relationship highlights that higher crude oil prices tend to cause the transportation and production costs which cause the food prices to increase leading to a significant rise in CFPI. In contrast, the correlation between repo rate and CFPI is -0.0096, which highlights an almost negligible linear relationship between the two variables. This suggests that although repo rate theoretically reduces inflationary pressures, its effect on food prices may be limited because food inflation is largely influenced by supply-side factors.

Implication and Significance:

This study provides an empirical assessment of the factors influencing food prices in India by examining the combined effects of fertilizer prices, Brent Oil Prices, USD/INR and EUR/INR exchange rates and the Reserve Bank of India's repo rate on CFPI. By analysing these determinants over the 2000-2025 period, the study provides a long term perspective on the relationship between domestic food prices and broader macroeconomic and agricultural conditions.

The study also contributes to existing literature by bringing agricultural input costs and macroeconomic variables together within a single regression framework. While previous have examined the chosen factors, this study evaluates their relationship with CFPI simultaneously which enables a more comprehensive assessment of which factors are statistically associated with food price movement in India.

Limitations:

The study has several limitations, The analysis is based on only 26 annual observations from 2000 to 2025, which limits the statistical power and generalisability of the findings. The study also identifies statistical associations rather than the casual relationships, Moreover, since the analysis uses annual observations and does not explicitly model lagged relationships, the dynamic effects of previous period food prices and explanatory variables are not captured. Future research could extend the analysis using larger datasets and dynamic time- series- specifications.

The scope of the study is limited to the selected variables and does not provide an exhaustive explanation of all determinants of food prices in India. Factors such as rainfall, temperature, agricultural production, crop yields, minimum support prices, food stocks, imports and exports, wages, money supply, government subsidies, and international food commodity prices are outside the primary scope of the model.

Closing Thoughts:

This research examined the factors affecting food prices inflation in India with the main objectives of investigating the relationship between Consumer Food Price Index(CFPI) and the selected macroeconomic and agricultural input-cost variables. By analysing annual secondary data from 2000-2025 through a multiple regression model, the sought to understand the extent to which each of the selected independent variables were associated with food price movements in India. The final findings indicate that fertilizer prices have the most significant positive relationship with CFPI while Brent Oil Prices, EUR/INR and USD/INR exchange rates have a moderately positive relationship and repo rate does not demonstrate statistically significant relationship with CFPI. Additionally, the study reveals that CFPI is more closely associated with supply-side input costs than monetary policy conditions, highlighting the importance of supply-side and external cost pressures.

The study contributes to the field of economic, particularly food prices price and inflation research by integrating agricultural input costs with international and domestic macroeconomic variables within a single empirical framework and offers practical insights for policymakers, agricultural authorities, economists and researchers.

In conclusions, the study demonstrates that food price movements in India are influenced by multiple economic forces, with fertilizer prices merging as a significant determinant, thereby highlighting the importance of agricultural input costs in understanding and addressing food price pressures.

References:

Adebayo, T. S. (2020). Dynamic relationship between oil price and inflation in oil exporting economy: Empirical evidence from wavelet coherence technique. *Energy Economics Letters, 7*(1), 12–22. https://doi.org/10.18488/journal.82.2020.71.12.22

Ahmed, M., & Singla, N. (2014). An analysis of major determinants of food inflation in India. *Indian Journal of Economics and Development, 10*(2), 173–182. https://doi.org/10.5958/2322-0430.2014.00547.2

Alexander, J., Arneth, A., Henry, R. C., Maire, J., Rabin, S., & Rounsevell, M. (2022). High energy and fertilizer prices are more damaging than food export curtailment from Ukraine and Russia for food prices, health and the environment. *Nature Food, 4*(1), 84–95. https://doi.org/10.1038/s43016-022-00659-

Alp, H., Klepacz, M., & Saxena, A. (2023). Second-round effects of oil prices on inflation in the advanced foreign economies. *FEDS Notes*. https://doi.org/10.17016/2380-7172.3401

Alviarez, V., Pedroni, P., Powell, A., & Quevedo Rocha, I. K. (2025). *International versus domestic shocks and pass-through to country prices: A heterogeneous VAR approach*. https://doi.org/10.18235/0013406

Bhattarai, S., Chatterjee, A., & Udupa, G. (2025). Food, fuel, and facts: Distributional effects of global price shocks. *International Finance Discussion Papers, 1414*. https://doi.org/10.17016/IFDP.2025.1414

Calvia, M. (2025). Dissecting extreme price fluctuations in mineral fertilizers: Regularities and co-movements in light of global food security. *Applied Economic Perspectives and Policy*. https://doi.org/10.1002/aepp.70010

Chandrasekhar, C. P. (2013). Not a benign market: An analysis of food price inflation and volatility. *Review of Market Integration, 2*(2), 121–159. https://doi.org/10.1177/2277976013493572

Chen, D., & Yang, R. (2024). The effects of energy commodities on CPI. *Journal of Student Research, 13*(2). https://doi.org/10.47611/jsrhs.v13i2.6753

Chen, S.-S. (2008). *Oil price pass-through into inflation*. Social Science Research Network.

Choi, S., Furceri, D., Furceri, D., Loungani, P., Mishra, S., & Poplawski-Ribeiro, M. (2017). *Oil prices and inflation dynamics: Evidence from advanced and developing economies*. Research Papers in Economics. https://ideas.repec.org/p/yon/wpaper/2017rwp-111.html

Diwakar, P. K. (2025). Seasonal variation in food inflation: A study of agricultural commodities in India. *Journal of Informatics Education and Research, 5*(1s). https://doi.org/10.52783/jier.v5i1s.3451

Fukač, M. (2011). Have rising oil prices become a greater threat to price stability? *Economic Review, 96*(4), 27–53.

Goldberg, L. S., & Campa, J. M. (2010). The sensitivity of the CPI to exchange rates: Distribution margins, imported inputs, and trade exposure. *The Review of Economics and Statistics, 92*(2), 392–407. https://doi.org/10.1162/REST.2010.11459

Gnutzmann, H., & Spiewanowski, P. (2016). *Fertilizer fuels food prices: Identification through the oil-gas spread*. Social Science Research Network. https://doi.org/10.2139/SSRN.2808381

Herve, O. (2012). *Fertilizer markets and their interplay with commodity and food prices* (Research Papers in Economics, No. 66). https://doi.org/10.2791/82136

Kumar, K. M., Chaturvedi, V., Nagaraj, M., & Hamdaan, M. (2010). Analysing the impact of RBI repo rate adjustments on inflation control in India (2010–2023). *Indian Scientific Journal of Research in Engineering and Management, 8*(12), 1–7. https://doi.org/10.55041/ijsrem39459

Kun, T., & Takács, I. (2023). The change in fertilizer prices due to the Russo-Ukraine War. *Connections: The Quarterly Journal, 22*(2), 91–104. https://doi.org/10.11610/connections.22.2.64

LeBlanc, M., & Chinn, M. D. (2004). Do high oil prices presage inflation? The evidence from G-5 countries. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.509262

Liu, Z. (2023). *The different impacts of a strong dollar on developed and emerging market economies*. https://doi.org/10.54097/hbem.v15i.9226

Mohamed, E. S., & Saleh, A. (2024). The impact of exchange rate change on inflation using bibliometric analysis from 1975 to 2024. *International Journal of Academic Research in Accounting, Finance and Management Sciences, 14*(4). https://doi.org/10.6007/IJARAFMS/v14-i4/23712

Renou-Maissant, P. (2019). Is oil price still driving inflation? *The Energy Journal, 40*(6). https://doi.org/10.5547/01956574.40.6.PREN

Suresh, N. (2012). Does the repo rate really control inflation? *Asian Journal of Research in Banking and Finance, 2*(3), 79–84. http://www.indianjournals.com/ijor.aspx?target=ijor:ajrbf&volume=2&issue=3&article=007

Understanding the nexus between international commodity prices and inflation in India: An econometric analysis. (2025). *Indian Journal of Economics and Development*, 446–456. https://doi.org/10.35716/ijed-23263

Varshini, B., Rakshan, M., Patil, V., & Kumar, K. M. (2024). Investigating the effectiveness of India's monetary policy in controlling inflation and maintaining price stability. *Indian Scientific Journal of Research in Engineering and Management, 8*(12), 1–7. https://doi.org/10.55041/ijsrem39577

Wibowo, H. E., Novanda, R. R., & Fauzi, A. (2023). Overview of the literature on the impact of food price volatility. *Agritropica: Journal of Agricultural Sciences, 6*(1), 22–32. https://doi.org/10.31186/j.agritropica.6.1.22-32

Yakimovich, E. A. (2023). Driving forces of global food crisis. *Vestnik of Astrakhan State Technical University: Series Economics, 2023*(3), 52–60. https://doi.org/10.24143/2073-5537-2023-3-52-60

An investigation into impact of repo rate on inflation indices (WPI & CPI) in India during pandemic and post pandemic phase. (n.d.). https://doi.org/10.52783/eel.v13i3.607

The inorganic fertilizer price surge in 2021: Key drivers and policy options. (2022). https://doi.org/10.22004/ag.econ.329241